Week 6

Question 1. The class boundary in a perceptron classifier is a:
A. Circle
B. Parabola
C. Straight line
D. Ellipse

A. Identify the Core Concept Being Tested The geometric interpretation of the decision surface generated by a single perceptron classifier in a 2-dimensional feature space.

B. Rigorous Solution The correct answer is **C. Straight line**.

C. Derivations / Mathematical Reasoning

1. The fundamental equation governing a perceptron's decision boundary is mathematically represented by setting the linear combination of inputs to zero: $\sum_{i=1}^{n} w_i x_i + w_0 = 0$, where w_i are the weights, x_i are the features, and w_0 is the bias.
2. In a 2-dimensional space ($n = 2$), the feature vector consists of x_1 and x_2.
3. Expanding the equation yields: $w_1 x_1 + w_2 x_2 + w_0 = 0$.
4. This equation follows the standard algebraic format of a linear equation ($Ax + By + C = 0$). Geometrically, this strictly defines a straight line in a 2D coordinate system.

E. Concept Expansion A single perceptron acts as a linear discriminant classifier. It partitions the continuous feature space into exactly two distinct half-spaces split by a linear hyperplane. In two dimensions, this hyperplane is a straight line; in three dimensions, it is a flat plane, and in higher dimensions, it is a hyper-plane. Due to this strict linear geometry, a single perceptron can only successfully classify datasets that are completely linearly separable.

Question 2 The number of neurons in a single perceptron classifier is?
A. 4
B. 3
C. 2
D. 1

A. Identify the Core Concept Being Tested The structural definition and composition of a standard single perceptron model.

B. Rigorous Solution The correct answer is **D. 1**.

C. Derivations / Mathematical Reasoning The perceptron algorithm is architecturally built upon a single biological-inspired artificial neuron. It acts as a single processing unit (threshold switching unit) that takes multiple weighted inputs, aggregates them via a summation junction, and applies a solitary activation function to produce a singular output. Therefore, a "single perceptron classifier" inherently consists of exactly 1 neuron.

E. Concept Expansion While a single perceptron uses one neuron, the theory of "Connectionism" posits that by connecting multiple simple units (neurons) together, networks can execute massively parallel, complex computations. A single perceptron is structurally the most basic building block of Artificial Neural Networks (ANNs), capable of processing simple linear boundaries, but multiple units must be connected to form Multi-Layered Perceptrons (MLPs) to achieve higher-order logic mapping.

Question 3 The MLP stands for:
A. Multi-Layered Perceptron
B. Mini-Layered Perceptron
C. Multi-Labelled Perceptron
D. Mini-Labelled Perceptron

A. Identify the Core Concept Being Tested Standard acronyms and nomenclature used in artificial neural network architectures.

B. Rigorous Solution The correct answer is **A. Multi-Layered Perceptron**.

C. Derivations / Mathematical Reasoning MLP is universally defined in the course material as a Multi-Layered Perceptron. It denotes an Artificial Neural Network topology where multiple layers of perceptron nodes (an input layer, one or more hidden layers, and an output layer) are cascaded together sequentially.

E. Concept Expansion The Multi-Layered Perceptron architecture was formulated to directly solve the limitations of the single perceptron. By cascading output signals from a primary perceptron layer into the inputs of subsequent hidden layers, MLPs can mathematically combine multiple linear boundaries. This creates complex, piece-wise non-linear decision boundaries required to classify non-linearly separable data.

Question 4 Which of the following logic functions can be implemented by a MLP, but cannot be implemented by a
single perceptron?
A. AND
B. OR
C. NOR
D. XOR

A. Identify the Core Concept Being Tested The limitation of linear separability in single perceptrons versus the capabilities of Multi-Layered Perceptrons for mapping Boolean logic.

B. Rigorous Solution The correct answer is **D. XOR**.

C. Derivations / Mathematical Reasoning

1. A single perceptron classifies by dividing a space with a single straight line.
2. For logical AND/OR, the true/false states group together, allowing a single line to successfully partition the 1s from the 0s.
3. For the XOR function, the truth table is: $(0,0) \to 0$, $(1,1) \to 0$, $(1,0) \to 1$, and $(0,1) \to 1$.
4. Plotting these four coordinates on a 2D graph reveals that the positive points (where XOR is 1) and negative points (where XOR is 0) lie on alternating diagonal corners.
5. No single straight line can physically be drawn to separate the positive group from the negative group without misclassifying at least one point.
6. Thus, a single perceptron cannot implement XOR. An MLP overcomes this by combining at least two linear bounds (generated by multiple nodes) via a hidden layer to encapsulate the target regions.

E. Concept Expansion This specific failure of the single perceptron to learn the XOR function was famously highlighted by Marvin Minsky and Seymour Papert in 1969. This discovery originally caused a massive decline in neural network research. It wasn't until the formulation of MLPs with hidden units and the backpropagation training algorithm that neural networks regained prominence, as the hidden units act as feature extractors capable of discovering new non-linear regularities.

Question 5 An MLP with a single hidden layer has 3 nodes in the input layer, 2 nodes in the hidden layer, and 1 node in the output layer. The total number of weights of the fully connected MLP is:

A. 8
B. 6
C. 4
D. 2

A. Identify the Core Concept Being Tested Calculating the total quantity of interconnecting synaptic weights in a feed-forward, fully connected Multi-Layered Perceptron.

B. Rigorous Solution The correct answer is **A. 8**.

D. Numerical Solution

- **Governing Equation:** In a fully connected feed-forward MLP, every node in a given layer L_i connects strictly to every node in the subsequent consecutive layer L_{i+1}. The number of weights connecting two layers is given by: $Weights = Nodes(L_i) \times Nodes(L_{i+1})$.

- **Substitute Values:**
  - Input layer (L_1) nodes = 3
  - Hidden layer (L_2) nodes = 2
  - Output layer (L_3) nodes = 1
- **Compute Intermediate Results:**
  - Weights between Input and Hidden Layer = $3 \times 2 = 6$.
  - Weights between Hidden and Output Layer = $2 \times 1 = 2$.
- **Final Result Interpretation:**
  - Total weights = $6 + 2 = 8$. A total of 8 trainable weights exist defining the full connection mapping of this network.

E. Concept Expansion This calculation represents the network's trainable weight vector dimensionality (excluding explicitly added bias weights). A fully connected network guarantees maximum data integration, ensuring that all feature combinations from the inputs are passed down into every hidden dimension. During training, this entire vector of 8 weights shapes the multi-dimensional hypothesis space over which the error gradient descent takes place.

Question 6 Which of the following is NOT true about the MLP?
A. It is a feed-forward network
B. It is a fully connected network
C. It is a neural network
D. It is a feed-back network

A. Identify the Core Concept Being Tested Architectural characteristics and data-flow directionality defining a standard Multi-Layered Perceptron (MLP).

B. Rigorous Solution The correct answer is **D. It is a feed-back network**.

C. Derivations / Mathematical Reasoning The options A, B, and C describe true properties of an MLP: it is an artificial neural network, it is a fully connected network, and it is explicitly a "feed forward network". In an MLP, signals strictly propagate forward from the input layer, through the hidden layers, to the output layer without any loops. Therefore, stating that an MLP is a "feed-back network" is mathematically and structurally false.

E. Concept Expansion A network that possesses loops where hidden units feedback to inputs, or previous layers, is categorized specifically as a "Recurrent Network" rather than an MLP. Recurrent networks maintain a form of memory representations of previous states for applications like time-series analysis, which fundamentally violates the pure, static, feed-forward topology of an MLP.

Question 7 A sigmoid activation function is commonly used in the MLP. The sigmoid activation function $AsigmoidactivationfunctioniscommonlyusedintheMLP.Thesigmoidactivationfunction$ h(z) = \frac{1}{1 + e^{-z}}$$is:

A. Discontinuous and not differentiable

B. Discontinuous but differentiable

C. Continuous but not differentiable

D. Continuous and differentiable

A. Identify the Core Concept Being Tested The mathematical properties of the sigmoid activation function that make it crucial for training neural networks.

B. Rigorous Solution The correct answer is **D. Continuous and differentiable**.

C. Derivations / Mathematical Reasoning

1. The classic perceptron utilizes a hard step/threshold function which makes a discontinuous jump from 0 to 1. At this vertical jump, the mathematical derivative is undefined (or infinite), and zero everywhere else.
2. The sigmoid function, defined as $h(z) = \frac{1}{1+e^{-z}}$, forms an "S-shape" that maps inputs smoothly between 0 and 1.
3. Because the curve lacks sharp jumps, it is inherently continuous.
4. It is everywhere differentiable. The derivative of the sigmoid $f(x)$ elegantly resolves to $f(x)(1 - f(x))$.
5. Therefore, the function is both strictly continuous and differentiable.

E. Concept Expansion Differentiability is an absolute requisite for training MLPs. Training requires calculating an error metric and propagating that error backward through the network to update weights, an operation mathematically reliant on calculating partial derivatives via the chain rule. Because step functions are non-differentiable, gradient descent search algorithms cannot navigate the weight space. Sigmoid units solve this by providing a continuous, non-linear, differentiable response.

Question 8 Which of the following step is used to update the weights of an MLP during the training phase:

A. Distillation

B. Backpropagation

C. Discretization

D. Elimination

A. Identify the Core Concept Being Tested The designated training algorithm utilized to update multi-layered neural network weights.

B. Rigorous Solution The correct answer is **B. Backpropagation**.

C. Derivations / Mathematical Reasoning While a single perceptron can use the simple Delta rule or Perceptron training rule because it directly observes the output error, Multi-Layered Perceptrons possess hidden units whose "correct" target values are inherently unknown. To assign blame for errors to these hidden units, the network computes the error at the output nodes and mathematically propagates those error terms *backwards* layer-by-layer through the network's connections. This process of updating weights via reverse error propagation is formally known as "backpropagation".

E. Concept Expansion The backpropagation procedure operates in two distinct phases:

1. **Forward Propagation:** The input vector is fed forward through the current weights to calculate the network's output and define the mean squared error.
2. **Backward Propagation:** The error derivative is passed backwards, dynamically computing the δ (delta) values for hidden units using the chain rule, scaling the necessary weight update for each specific neuron iteratively until convergence.

Question 9 Which of the following optimization technique is implemented by backpropagation during neural
network training?
A. Quadratic programming
B. Linear programming
C. Gradient descent
D. Evolutionary algorithms

A. Identify the Core Concept Being Tested The mathematical optimization strategy executed inside the backpropagation algorithm.

B. Rigorous Solution The correct answer is **C. Gradient descent**.

C. Derivations / Mathematical Reasoning

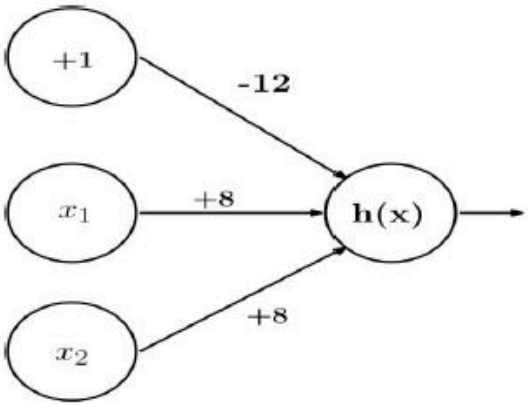
1. To train a neural network, one must minimize the error function $E = \frac{1}{2} \sum (t_d - o_d)^2$.
2. The error surface spans an n-dimensional weight space.
3. To reach the minimum error (the bottom of the surface), the backpropagation algorithm calculates the derivative (gradient) of the error surface concerning each weight: ∇E.
4. It dynamically alters the weight vector in the exact inverse direction of the gradient: $\Delta w = -\eta \nabla E$.
5. This mechanism of mathematically sliding down the steepest slope of a multi-dimensional function to find a minimum is formally termed "Gradient descent".

E. Concept Expansion Gradient descent contrasts starkly with algorithms like Quadratic Programming (QP). While Support Vector Machines (SVMs) utilize QP to find a guaranteed global minimum for convex problems, gradient descent in highly complex MLPs may contain multiple localized valleys. Consequently, standard backpropagation using gradient descent is

susceptible to getting stuck in "local minima" rather than finding the true global minimum. Techniques like adding momentum are used to jump out of local traps.

Question 10

Consider the neural network given below. It takes two binary valued inputs $x_1, x_2 \epsilon \{0,1\}$ and the activation function is the binary threshold function ($h(x) = 1 \; if \; x > 0; 0 \; otherwise$). What is the output of the neural network for the input vector [1, 1]?



A. 0

B. 1

C. 4

D. 16

A. Identify the Core Concept Being Tested The mathematical forward-propagation of inputs through a neural network's linear combination equation and activation function.

B. Rigorous Solution The correct answer is **B. 1**.

D. Numerical Solution

- **Governing Equation:** The output of a neuron is given by evaluating the activation function $h(net)$ where the internal linear combination is $net = w_0 x_0 + \sum_{i=1}^{n} w_i x_i$.
- **Substitute Values:**
  - The bias node intrinsically supplies an input of $x_0 = 1$ with a weight of $w_0 = -12$.
  - The problem specifies the input vector is , therefore $x_1 = 1$ and $x_2 = 1$.
  - The network diagram specifies weights $w_1 = +8$ and $w_2 = +8$.
- **Compute Intermediate Results:**
  - $net = (1 \times -12) + (1 \times 8) + (1 \times 8)$
  - $net = -12 + 8 + 8 = 4$
- **Apply Activation:**
  - The problem states the binary threshold function is: $h(x) = 1$ if $x > 0$; 0 otherwise.
  - Since $net = 4$, and $4 > 0$, $h(4) = 1$.

- **Final Result Interpretation:** The linear discriminant outputs a positive value of 4, indicating the point falls strictly on the positive side of the threshold boundary. Therefore, the discrete output of the neural network is 1.

E. Concept Expansion This specific neural network structure maps exactly to the Boolean logical **AND** gate. An AND gate only triggers when both inputs x_1 and x_2 are precisely 1. Only the input condition generates a sum of 16, successfully overcoming the heavy -12 bias threshold to output a 1. Any other binary state (,) will yield a linear combination of 0, -4, or -4 respectively, thereby failing to break the 0 threshold and outputting 0.

Question 11 When do you expect overfitting to happen in the following scenarios of neural network training
iterations?
A. training set as well as test set error decreases
B. training set error decreases but test set error increases
C. training set error as well as test set error increases
D. training set as well as test set error remains constant

A. Identify the Core Concept Being Tested Detecting "overfitting" by comparatively analyzing training error and validation/test error curves over learning epochs.

B. Rigorous Solution The correct answer is **B. training set error decreases but test set error increases**.

C. Derivations / Mathematical Reasoning

1. Neural networks train by performing gradient descent solely on the *training set* data. Over successive iterations, the algorithm will continuously map decision boundaries tighter to the training samples, guaranteeing that the **training error strictly decreases**.
2. However, if the network topology is too complex or trained for too many iterations, it begins to memorize idiosyncratic noise in the training set rather than learning true underlying patterns.
3. When this happens, the network loses its ability to generalize to new, unseen data. If tested concurrently on an external validation/test set, the **test set error will begin to rise** simultaneously as the training error keeps falling.

E. Concept Expansion Overfitting poses a fundamental trade-off between classifier complexity and generalization error. To prevent catastrophic overfitting, practitioners typically deploy a "validation set". They monitor the error plots simultaneously and employ "early stopping"—halting the training sequence explicitly at the trough where validation error reaches its minimum, right before it begins expanding upward due to over-complex mapping.